VIT-AP UNIVERSITY

Project Report

Mentor: Dr. Ravisankar Barpanda

Project Title: Hand Gestures Controlled Car

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1. Identification of the Problem and Project Title

- **Project Title:** Wireless Gesture Controlled Car
- **Problem Identification:** The primary objective is to develop a car that can be wirelessly controlled using hand gestures. This system will be particularly beneficial for individuals with limited mobility, offering them an easy way to control movement without using traditional input devices such as a joystick or phone. This can also be used to test hand stability in various medical fields such as in Medical Colleges to train and test a medical student hand stability. We have also developed Obstacle detection and Avoidance feature for safety.

2. Identification of Required Components

Upon our research on this project, we found that these are the tentative components that are needed for the completion of the project:

- 4 Wheel Robot Smart Car Chassis Kit
- HC-05 Bluetooth Module
- Arduino Nano Power Supply
- Two HC-05 Bluetooth modules
- gy-521 MPU 6050 Module
- L298n Motor Driver
- 18650 9V four batteries
- Two 18650 2-cell battery holders
- DC Power Switch
- Jumper Wires
- Servo Motor
- Ultrasonic Sensor

3. Actual Cost

S.No	Component Name	Estimated Cost(in Rs)
1	4 Wheel Robot Smart Car Chassis Kit	480
2	Arduino UNO	460
3	Arduino Nano	220
4	Two HC-05 Bluetooth modules	220 x 2 = 440
5	gy-521 MPU 6050 Module	140
6	L298n Motor Driver	130
7	18650 9V four batteries	280
8	Two 18650 2-cell battery holders	30 x 2 = 60
9	DC Power Switch	5
10	Jumper Wires	15
11	Servo Motor	80
12	Ultrasonic	260

Total Budget: Rs. 2590/- (depending on availability and quality of parts.)

4. Plan of Action

- **Research & Component Procurement:**
  - Study each component, its functionality, and how it integrates into the project.
  - Purchase necessary components.
- **Hardware Setup:**
  - Set up the Arduino connections, including motors and accelerometer.
  - Install the Bluetooth modules and establish wireless communication.
- **Software Development:**
  - Write code to control the car using the accelerometer's data.
  - Implement gesture recognition algorithms for forward, backward, left, and right motion.
- **Testing & Debugging:**
  - Test the wireless gesture control functionality and ensure smooth data transmission.
  - Debug and troubleshoot issues with motor control and Bluetooth connectivity.

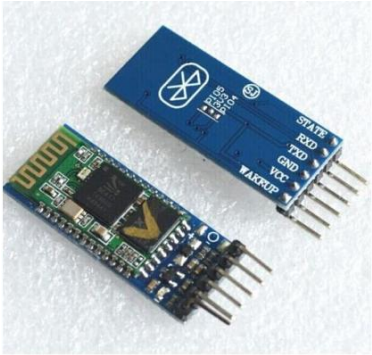
- **Final Implementation:**
  - Fine-tune the system and demonstrate the working model.

5. Timeline of Actions

- ✓ **Week 1:** Research and procure components.
- ✓ **Week 2:** Set up the hardware and connect the components.
- ✓ **Week 3-4:** Start writing the code for gesture recognition and wireless communication.
- ✓ **Week 5-6:** Test the system for basic functionality (forward, backward, left, right movements and Obstacle avoidance).
- ✓ **Week 7:** Debug any hardware/software issues and refine the system and Final testing and demonstration.

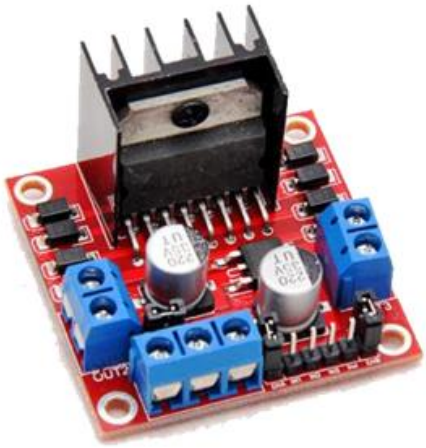
About Components:

1. HC 05 Bluetooth Module:



The HC-05 Bluetooth Module has 6 pins- VCC, GND, TX, RX, Key, and LED. It comes pre-programmed as a slave, so there is no need to connect the Key pin, unless you need it change it to Master Mode. The major difference between Master and Slave modes is that, in Slave mode the Bluetooth module cannot initiate a connection, it can however accept incoming connections. After the connection is established the Bluetooth module can transmit and receive data regardless of the mode it is running in. If you are using a phone to connect to the Bluetooth module, you can simply use it in the Slave mode. The default data transmission rate is 9600kbps. The range for Bluetooth communication is usually 30m or less. The module has a factory set pin of “1234” which is used while pairing the module to a phone.

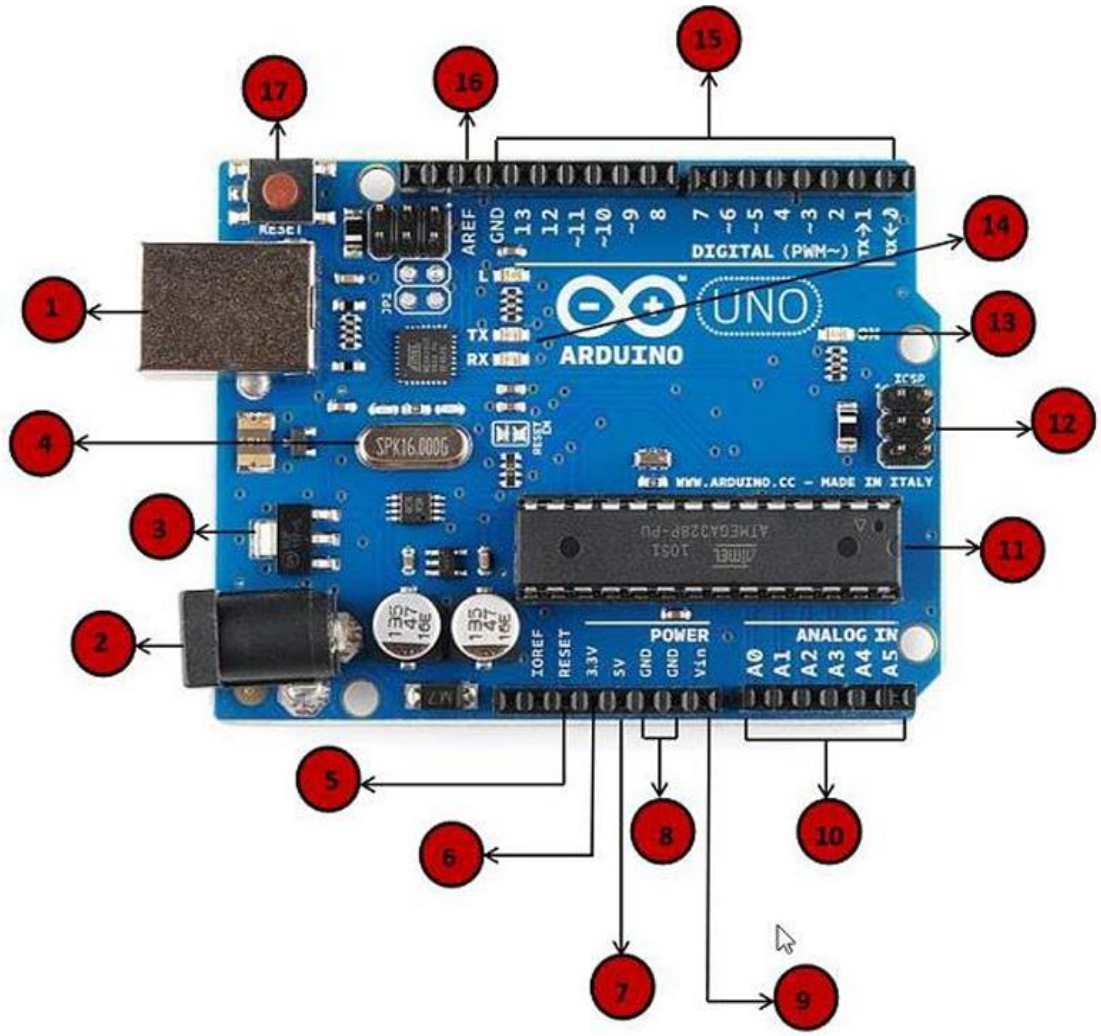
2. L298D Motor drivers:



L298D is a dual H-bridge motor driver integrated circuit (IC). Motor drivers act as current amplifiers since they take a low-current control signal and provide a higher-current signal. This higher current signal is used to drive the motors. L298D contains two inbuilt H-bridge driver circuits. In its common mode of operation, two DC motors can be driven simultaneously, both in forward and reverse direction. The motor operations of two motors can be controlled by input logic at pins 2 & 7 and 10 & 15. Input logic 00 or 11 will stop the corresponding motor. Logic 01 and 10 will rotate it in clockwise and anticlockwise directions, respectively. Enable pins 1 and 9 (corresponding to the two motors) must be high for motors to start operating. When an enable input is high, the associated driver gets enabled. As a result, the outputs become active and work in phase with their inputs. Similarly, when the enable input is low, that driver is disabled, and their outputs are off and in the high-impedance state.

3. Arduino (UNO):

Arduino UNO board is the most popular board in the Arduino board family. In addition, it is the best board to get started with electronics and coding. Some boards look a bit different from the one given below, but most Arduinos have majority of these components in common.



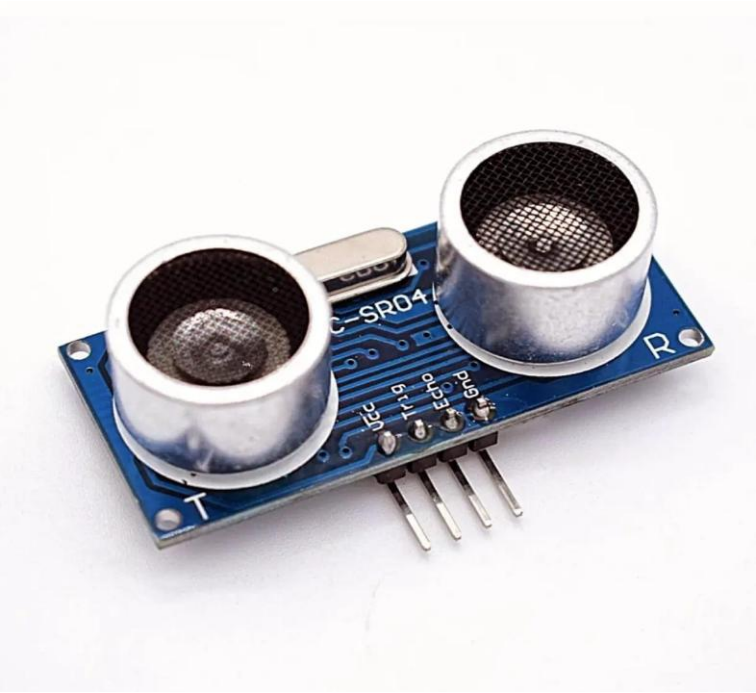
1	**Power USB** Arduino board can be powered by using the USB cable from your computer. All you need to do is connect the USB cable to the USB connection (1).
2	**Power (Barrel Jack)** Arduino boards can be powered directly from the AC mains power supply by connecting it to the Barrel Jack (2).
3	**Voltage Regulator** The function of the voltage regulator is to control the voltage given to the Arduino board and stabilize the DC voltages used by the processor and other elements.
4	**Crystal Oscillator** The crystal oscillator helps Arduino in dealing with time issues. How does Arduino calculate time? The answer is, by using the crystal oscillator. The number printed on top of the Arduino crystal is 16.000H9H. It tells us that the frequency is 16,000,000 Hertz or 16 MHz.
5 , 17	**Arduino Reset**

	You can reset your Arduino board, i.e., start your program from the beginning. You can reset the UNO board in two ways. First, by using the reset button (17) on the board. Second, you can connect an external reset button to the Arduino pin labelled RESET (5).
6, 7, 8, 9	**Pins (3.3, 5, GND, Vin)** • 3.3V (6) − Supply 3.3 output volt • 5V (7) − Supply 5 output volt • Most of the components used with Arduino board works fine with 3.3 volt and 5 volt. • GND (8)(Ground) − There are several GND pins on the Arduino, any of which can be used to ground your circuit. • Vin (9) − This pin also can be used to power the Arduino board from an external power source, like AC mains power supply.
10	**Analog pins** The Arduino UNO board has five analog input pins A0 through A5. These pins can read the signal from an analog sensor like the humidity sensor or temperature sensor and convert it into a digital value that can be read by the microprocessor.
11	**Main microcontroller** Each Arduino board has its own microcontroller (11). You can assume it as the brain of your board. The main IC (integrated circuit) on the Arduino is slightly different from board to board. The microcontrollers are usually of the ATMEL Company. You must know what IC your board has before loading up a new program from the Arduino IDE. This information is available on the top of the IC. For more details about the IC construction and functions, you can refer to the data sheet.
12	**ICSP pin** Mostly, ICSP (12) is an AVR, a tiny programming header for the Arduino consisting of MOSI, MISO, SCK, RESET, VCC, and GND. It is often referred to as an SPI (Serial Peripheral Interface), which could be considered as an "expansion" of the output. Actually, you are slaving the output device to the master of the SPI bus.
13	**Power LED indicator** This LED should light up when you plug your Arduino into a power source to indicate that your board is powered up correctly. If this light does not turn on, then there is something wrong with the connection.
14	**TX and RX LEDs** On your board, you will find two labels: TX (transmit) and RX (receive). They appear in two places on the Arduino UNO board. First, at the digital pins 0 and 1, to indicate the pins responsible for serial communication. Second, the TX and RX led (13). The TX led

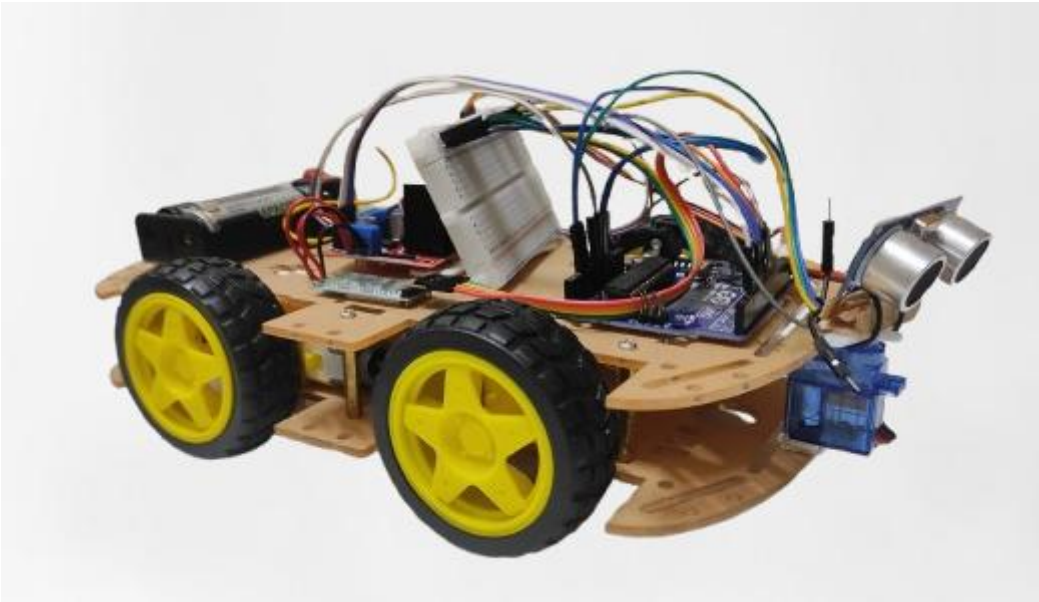
	flashes with different speed while sending the serial data. The speed of flashing depends on the baud rate used by the board. RX flashes during the receiving process.
15	**Digital I/O** The Arduino UNO board has 14 digital I/O pins (15) (of which 6 provide PWM (Pulse Width Modulation) output. These pins can be configured to work as input digital pins to read logic values (0 or 1) or as digital output pins to drive different modules like LEDs, relays, etc. The pins labeled "~" can be used to generate PWM.
16	**AREF** AREF stands for Analog Reference. It is sometimes, used to set an external reference voltage (between 0 and 5 Volts) as the upper limit for the analog input pins

4. Ultrasonic Sensor:

An ultrasonic sensor is a device that uses sound waves to detect the distance to an object. It operates by emitting a high-frequency ultrasonic pulse through a transmitter and waiting for it to bounce back to the receiver. By measuring the time taken for the sound wave to travel to the object and return, the sensor calculates the distance based on the speed of sound in air. Ultrasonic sensors typically operate with trigger and echo pins. When a high pulse is sent to the trigger pin, the sensor transmits an ultrasonic wave. Once the wave reflects back, it is detected at the echo pin, which remains high for a duration proportional to the distance of the object. Ultrasonic sensors are commonly used for distance measurement and obstacle detection in robotics and automation projects, as they provide accurate readings in a non-contact manner and are unaffected by surface colour or transparency.



Hand Gestured Controlled Car:

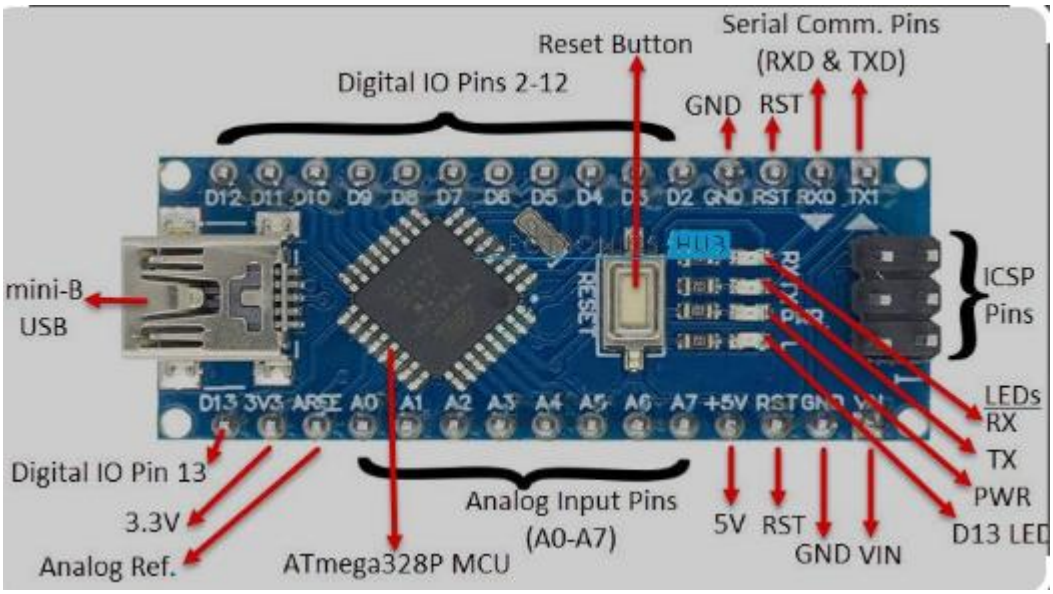


Hand controller:

Advantages:

1. **Intuitive Interaction:** Gestures are natural and easy to learn, making it simple for users to interact with devices without needing complex input methods.
2. **Enhanced User Experience:** Controlling a car with hand gestures can provide an intuitive and futuristic experience, allowing drivers to manage certain functions with simple movements instead of physically pressing buttons.
3. **Reduced Distractions:** Hand gestures can allow drivers to keep their eyes on the road while adjusting certain settings, which could reduce the risk of accidents associated with looking down to locate buttons or touchscreens.
4. **Improved Accessibility:** For individuals with physical disabilities or limited mobility, hand gestures can provide an alternative control method that doesn't rely on fine motor skills or reaching for controls.
5. **Increased Efficiency**: Quick, simple gestures can often perform tasks faster than traditional input methods, streamlining user interaction
6. **Increased Safety:** Some gestures may allow for more rapid reactions, like a quick wave to control the stereo or adjust climate settings, which could minimize the time needed for adjustments.
7. **Contactless Control**: Ideal for situations where hygiene is important, such as medical environments, public kiosks, or industrial settings, minimizing the need for direct contact.
8. **Greater Customization:** Gesture control systems can be programmed to suit individual driver preferences, potentially improving user satisfaction and comfort.
9. **Reduced Wear and Tear on Physical Controls:** By minimizing physical contact with buttons and controls, gesture controls can help reduce the wear on interior components, potentially lowering maintenance costs over time.
10. **Integration with Smart Systems:** Gesture control can integrate with other smart vehicle systems, such as voice recognition and AI, for a seamless and adaptive in-car experience.

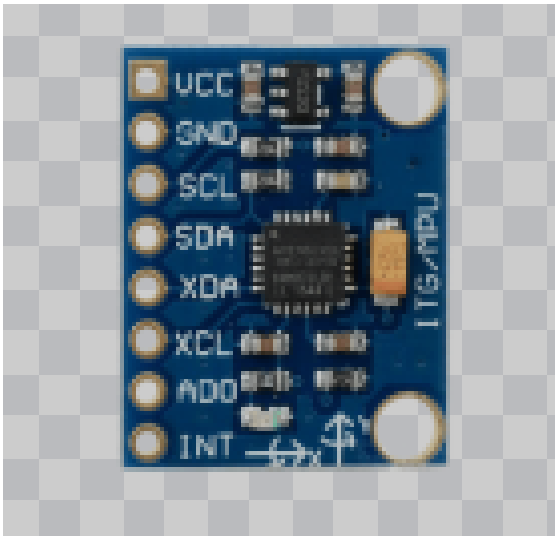
4. Arduino (NANO): The **Arduino Nano** is a small, compact microcontroller board based on the **ATmega328P** (same as the Arduino Uno) with built-in USB support for easy programming.



 Microcontroller (ATmega328P) – The main chip that runs the code.
 USB Connector – Used for programming and powering the Nano.
 Digital Pins (D0 to D13) – 14 digital input/output pins (for reading sensors or controlling components).
 Analog Pins (A0 to A7) – 8 analog input pins (for reading analog sensors).
 Power Pins – Includes VCC (5V), GND (Ground), and other power-related pins like RAW (for external power supply).
 Reset Button – Used to reset the microcontroller.
 Status LEDs – For power and digital pin 13 (used for debugging).
 Crystal Oscillator – Provides the clock for the microcontroller (usually 16 MHz).
 Programming Port (TX, RX) – For serial communication with your computer.

5.MPU 6050

The **MPU-6050** is a popular **6-axis motion tracking** device, often used in Arduino and other embedded systems projects. It combines a **3-axis gyroscope** and a **3-axis accelerometer** in one chip, allowing for the measurement of both **angular velocity** and **linear acceleration** in three-dimensional space.



Key Pins on the MPU-6050:

VCC	Power supply pin (typically 3.3V or 5V depending on your board)
GND	Ground pin
SCL	Serial Clock Line (I2C communication)
SDA	Serial Data Line (I2C communication)
INT	Interrupt pin (can be used to trigger events based on motion detection)

Key Features of the MPU-6050:

- **3-axis accelerometer** (measures acceleration along X, Y, and Z axes)
- Range options: ±2g, ±4g, ±8g, and ±16g
- **3-axis gyroscope** (measures angular velocity along X, Y, and Z axes)
  Range options: ±250°, ±500°, ±1000°, and ±2000° per second
- **Built-in Digital Motion Processor (DMP)** for advanced processing
- **I2C interface** for easy communication with microcontrollers (such as Arduino)

Applications:

- **Motion sensing**: Detects orientation, acceleration, and rotational movements.
- **Inertial navigation systems**: Used in robotics, drones, and wearable tech.
- **Game controllers and VR**: Tracks movements to interact with virtual environments.
- **Stabilization systems**: For balancing and stabilizing devices like drones and camera gimbals.